

Sudip Roy

Full-Stack Software Engineer

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Summary

Full-stack engineer who gets genuinely excited about building things. I built and run a platform for a Discord community that grew to 60,000+ members and 4,000+ daily users. The stack wanders across Python, Rust, and TypeScript because I kept hitting walls and learning new tools to break through them. I own the whole thing end to end: architecture, databases, deploys, and the 3 AM production debugging sessions.

Technical Skills

Languages: Python, Rust, TypeScript, SQL, JavaScript

Frameworks & Tools: Next.js, React, Axum, Actix, gRPC, discord.py, Docker, Git, GitHub Actions

Databases & Caching: PostgreSQL, Redis, MongoDB, SQLite, libSQL

Infrastructure: AWS, Cloudflare, Linux, static site deployment, service-oriented architecture

Experience

Software Developer

2020 – Present

Worldwide Discord Platform

Full Stack / Systems

- Built and maintained a production community platform serving 60,000+ members and 4,000+ daily active users across Python, Rust, and TypeScript services.
- Designed a three-layer architecture: Python handles the Discord gateway, Rust owns the data and financial path, and Next.js powers the dashboard and admin workflows.
- Introduced gRPC between services to separate bot logic from database and transaction logic, which made the codebase way easier to reason about.
- Evolved storage from raw JSON to MongoDB, Redis, PostgreSQL, and embedded libSQL. Each migration was triggered by a real production failure I had to fix.
- Embedded libSQL inside the Rust process to kill network round-trips for latency-sensitive operations, which noticeably improved the transaction path.
- Implemented economy systems, moderation tooling, games, event workflows, and real-money purchases with Axum and PayPal webhooks. Broke things, fixed them, added logging and recovery so they wouldn't break the same way twice.

Projects

Pattern Pursuit

2023

github.com/drlove2002/pattern_pursuit

Rust, TypeScript, Redis, Docker, AWS

- An interactive prediction game with a Rust/Actix backend, TypeScript frontend, and Redis for real-time game state.
- Uses 5-gram statistical language modeling so players compete against a model that learns from their gameplay.
- Containerized with Docker and deployed on AWS.

Sudoku Solver Research

Active

github.com/drlove2002/sudoku_solver

Rust, Criterion, Algorithm Research

- A modular Sudoku-solving and benchmarking system in Rust for testing swappable solver components against a large real-world dataset.
- Focused on minigrid-based constraint propagation, permutation graph construction, benchmarking, and graph-based visualization rather than a single hardcoded solver path.

Education

M.Sc. in Computer Science

Final Year

University of Calcutta

West Bengal, India

B.Sc. in Computer Science

2020 – 2023

University of Calcutta

West Bengal, India

Additional

Focus Areas: distributed systems, backend engineering, API design, database architecture, systems reliability, full-stack product development

Languages: Bengali (Native), English (Fluent), Hindi (Intermediate)

May 2, 2026

Dear Hiring Manager,

I'm gonna be real with you. I don't code for money. I mean yeah, I need food, clothes, a roof. The basics. That's it. Everything else? I just wanna build stuff, break it, fix it, learn something new, and do it all over again the next day. It's not a job for me. It's legit an addiction. A good one, I think.

My journey started like a low level character in an old MMO. I joined this dead Discord server called immunity. No one talked. I thought, what if I build a bot to wake people up? I barely knew Python. But instead of watching tutorials forever, I just... started. That's how I am. I get excited and I can't wait.

First bot was ugly. I added an economy system where people earned virtual coins for each message. It worked kinda. Then concurrent writes crashed it because I used a raw JSON file. Stupid mistake. But I loved every second.

I hopped into databases. MongoDB sharded cluster for fast writes. Then Redis for caching when things got slow. Then PostgreSQL when I needed relations. Each time I learned just enough to make it work, broke something else, then learned more. That cycle is my happy place.

Then I got obsessed with Rust. I built a Data Access Layer with gRPC so my Python bot could talk to Rust fast. Added an Axum API for real money purchases (people actually bought stuff). Then libSQL inside Rust to kill network lag. Every new thing felt like discovering a secret dungeon.

But here's what I'm really excited about lately. AI. I can't stop reading about new AI threads. I'm customizing my own AI agent using Pi mono. Yeah it's still janky but I love it. I'm building workflows that help me code my Discord bot and website faster. Stuff like auto generating boilerplate, summarizing logs, even suggesting refactors. It feels like I have a little wizard helper now.

And I'm also building new games. I can't talk about them yet. Seriously, I promised. But they're gonna be something. The idea is to create a feedback loop with the community. Like, an ultimate community that works buttery smooth. Money flows in booooo... well, at least it'll make my co-owner Flame from the Discord community happy XD. That's all I care about. Flame has been there since day one. If I can make him smile, I'm good.

And I'm not gonna lie. I get super excited about everything I build. Like, child level excitement. Dancing in my chair excited. I don't need a big salary. Just enough to survive and keep doing this forever. The rest of my life I wanna spend understanding software, breaking it, rebuilding it, and maybe helping some people along the way.

So yeah. I'm not a perfect developer. I misspell things. Sometimes I forget to commit for days. But I love coding with my whole dumb heart. If your company needs someone who genuinely does this for the love of the game, not for the incentives, I'm your guy.

Sincerely,
Sudip Roy